

NestureAI

Primitive Reflex Detection System

How the patterns are defined, how MediaPipe observes them, why each game targets the number it does, and what reaches the administrator export.

SCOPE	TAXONOMY	CAPTURE
8 games 2 input modes	8 primitive reflexes 3 function measures	3 MediaPipe graphs up to 520 points/frame

This document describes movement patterns to guide practice. It is not a medical assessment and does not identify any condition.

1 What a primitive reflex is

A primitive reflex is an automatic, whole-body movement pattern driven from the brainstem. It is normal and useful in an infant, and it is expected to **integrate** - to stop firing - as higher brain centres mature and take over voluntary control. Finding one still active in a school-age child is the finding: it means a movement the child must consciously override is still competing with the movement they intend.

Why that matters for this platform

A retained reflex shows up as effort in the wrong place. A child with an active ATNR who turns their head to look at a keyboard gets an involuntary arm extension on the same side; to point accurately they must fight their own arm. The games in this platform are built so that ordinary play repeatedly loads those patterns - reaching across the midline, holding one finger out while the rest of the hand stays quiet, keeping posture while the head moves - and the camera records what the body does while it happens.

Two different things were being called 'reflexes'

The platform separates two families, because they are read in opposite directions and mixing them produced a real reporting fault. Both are scored on the same 0-100 axis - *how much of a problem is there* - but that axis means retention in one family and dysfunction in the other.

Family	Members	A high score means	Reported vocabulary
Primitive reflexes Brainstem patterns that should integrate with maturity	ATNR, STNR, TLR, Moro, Palmar Grasp, Babkin, Hand-to-Mouth, Spinal Galant	The pattern is still active - it has not integrated	Clearly active / Partly active / Mostly settled / Settled
Function measures Lifelong reflexes and oculomotor skills	VOR (vestibulo-ocular reflex), Eye Coordination, Visual Tracking	The response is working poorly - it should be strong at every age	Needs support / Developing / Working well

The vocabulary fault this separation fixes

The vestibulo-ocular reflex never integrates - a weak one is a deficit, not a retention. When both families shared one set of words (strong / moderate / weak), a child whose VOR was *failing* was printed as "**VOR: strong**". Next to "retained reflex" that reads as bad news; next to "Vestibulo-Ocular Reflex" it reads as good news. Same word, opposite meaning, same report.

The underlying numbers were always correct - `vor.js` maps poor VOR quality to a high score deliberately. Only the words were wrong, so each family now gets its own vocabulary, shown in the last column above.

One reflex is never inferred from gameplay

Spinal Galant is elicited by stroking the skin alongside the spine. A webcam pointed at a seated child's hands and face cannot observe it under any circumstances. It is marked questionnaire-only: it is recorded from the intake form or a practitioner's observation, and is never inferred from cursor wobble or 'erratic control', which have a hundred other causes.

2 The patterns, and what each one looks like

Eight primitive reflexes and three function measures are modelled. The third column is the observable consequence the detector is actually looking for.

Pattern	What it is	What the camera can see
ATNR Asymmetrical Tonic Neck	Turning the head to one side drives the arm on that side to extend and the opposite arm to bend.	Head yaw correlates positively with asymmetric index-finger extension between the two hands.
STNR Symmetrical Tonic Neck	Head tilt up or down drives a symmetric change in arm and leg tone.	Head pitch correlates with the mean index extension of both hands together.
TLR Tonic Labyrinthine	Head position relative to gravity changes whole-body extension or flexion tone.	Head pitch correlates with overall hand-opening across both hands.
Moro	The infant startle: arms fling outward, then draw back in.	A simultaneous bilateral outward burst of both hands followed by an inward return.
Palmar Grasp	Pressure or effort in the palm pulls the fingers into a closed fist.	Rapid decrease in finger-to-wrist distance during effort, plus response speed and left/right symmetry.
Babkin	Pressure in the palms triggers mouth opening - a hand/mouth coupling.	Bilateral hand closure followed by mouth opening inside a 300-800 ms window.
Hand-to-Mouth rooting proxy	The hand travels to the mouth and the mouth prepares to receive it.	Hand-to-mouth distance, approach speed, mouth opening in response, and head/hand alignment.
Spinal Galant	Stroking beside the spine causes the hip to swing toward that side.	Not observable by camera. Questionnaire or practitioner only.

Function measures (not primitive reflexes)

Measure	What it is	What the camera can see
VOR	Vestibulo-ocular reflex: the eyes counter-rotate to hold an image still while the head moves. Lifelong.	Inverse correlation between head movement and iris movement. A healthy VOR sits near -1.
Eye Coordination	The two eyes working as a pair. An oculomotor skill, not a reflex.	Relative iris positions within the face mesh across the session.
Visual Tracking	Smooth pursuit: following a moving target with the eyes.	Iris path against the on-screen target and the hand that follows it.

3 How MediaPipe captures the points

Three different MediaPipe graphs run in this platform. Which one is running decides which reflexes can be measured at all, so this is the single most important table in the document.

Graph	Points per frame	Rate / resolution	Where it runs	Feeds
MediaPipe Hands	21 per hand up to 2 hands = 42	~30 fps 640 x 480 complexity 0 or 1	Camera mode in every game	Moro, Palmar Grasp, and the hand half of every other detector
Hands + Face Mesh with iris refinement	478 face (468 mesh + 10 iris) + 42 hand = 520	~30 fps 640 x 360 2 hands, 1 face	LetterQuest and Path Tracing only	Head pose, mouth aperture, iris - so ATNR, STNR, TLR, Babkin, Hand-to-Mouth, VOR, eye measures
MediaPipe Pose upper body	33 returned, 21 recorded (head to hips)	15 Hz sampled 640 x 480 complexity 1	Touch mode , with consent	Posture research indicators - a separate, provisional channel

Camera mode ('hand in the air') versus Touch mode

The two input modes are not two settings on one pipeline - they run different models for different purposes, and they are mutually exclusive. In every game the pose capture is gated on the exact complement of the hand tracker's condition, so two MediaPipe graphs can never compete for the same webcam.

	Camera mode - hand in the air	Touch mode
What drives the game	The index fingertip becomes the pointer. Selection is by dwell: hold on a target and a ring fills.	Finger or mouse on the screen. The camera drives nothing.
Model running	Hands (all games), plus Face Mesh in LetterQuest and Path Tracing.	Pose only, and only after consent. The hand tracker is off.
Reflex measurement	Full reflex engine: 5-second windows, 30 fps, all detectors the sensor set allows.	No primitive-reflex scoring. The report says so: "This round was played by touch, so the camera never ran."
What is stored	Per-window reflex results, plus raw hand points in <code>raw_hand_tracking</code> .	Upper-body landmarks in <code>raw_pose_tracking</code> , plus derived features and provisional indicators.
Consent	The camera is the game; consent is part of signing up for camera play.	Asked every round, before the camera opens. Declining changes nothing about the game.

Why touch mode records pose at all, and why it is kept separate

Touch mode has always meant the camera is off. Turning it on to observe posture is a change the child is asked about explicitly, every round, in words a child and a parent can both read. No video or image is ever stored - only the coordinates of 21 body points.

These pose indicators are **provisional and unvalidated**. The model is versioned `pose-indicators-v1.0.0-provisional`, and every reflex weight in its scoring table is deliberately **zero**: a child must never be marked down by a research number nobody has calibrated yet. Only directly observed postural measures (sitting steadiness, shoulder symmetry, midline crossing) carry any weight, capped at **15%** of a game score.

4 The detection rules

Each detector turns a window of landmark frames into a 0-100 score. The rules below are the actual constants in the code, not illustrative numbers.

How a session is analysed

Stage	Rule	Why
Window	150 frames (~5 s at 30 fps). Each window is analysed as it completes; only its result is kept.	A 300-frame ring buffer would have scored a ten-minute session on its last ten seconds. On one test session Palmar Grasp fell from 84 to 20 purely because the child sat still while the results loaded.
Minimum data	5 frames minimum for any analysis; a trailing partial window under 30 frames is discarded.	A fragment of a window is noise, not a shorter measurement.
Session result	Confidence-weighted aggregate of every completed window.	A reflex measurable in 6 windows out of 60 is a weaker claim than one measurable throughout, and the confidence carries that.
No data	Reported as not_measured - never as <i>none</i> with a score of 0.	"None" means measured and integrated. A camera that saw nothing once produced a clean bill of health and sent a score of 100 for a reflex nobody had observed.

Score to label

Score	Label	Primitive reflex reads as	Function measure reads as
65 - 100	strong	Clearly active - pattern showed up often	Needs support - responded weakly
40 - 64	moderate	Partly active - showed up some of the time	Developing - present, not yet consistent
20 - 39	weak	Mostly settled - only occasional	Working well
0 - 19	none	Settled - did not show up	Working well

A result is only surfaced as a finding when its confidence reaches **0.35**. Below that it is held back rather than printed, because a low-confidence "strong" is the most misleading thing a report can say. Confidence is banded as High at 0.70 and above, Medium from 0.35.

Named thresholds inside the detectors

Detector	Constant	Value	Meaning
Babkin	BABKIN_MIN_MS / MAX_MS	300 / 800 ms	The window in which mouth opening must follow hand closure to count as coupled
Babkin	HAND_CLOSE_THRESHOLD	0.35	Hand-open score below this counts as a closed hand
Babkin	MOUTH_OPEN_THRESHOLD	0.02	Normalised lip separation that counts as an open mouth
Palmar Grasp	CLOSE_THRESHOLD	0.05 / s	Rate of finger-to-wrist shortening that counts as a rapid closure
Hand-to-Mouth	APPROACH_THRESHOLD	0.30	Normalised hand-to-mouth distance under which the hand counts as 'near'
Hand-to-Mouth	MOUTH_REACT_THRESHOLD	0.025	Minimum mouth opening that counts as the mouth reacting
Moro	MIN_BILATERAL_FRAMES	15	Frames with both hands visible before a startle shape can be claimed

5 Why a game lists only two reflexes

This is the question the per-game numbers raise, and the answer is a deliberate design rule rather than an omission. Two different claims are being made, and they pass through two different gates.

Claim	What it means	When it is true
TARGET	What the activity is designed to work on.	A clinical statement about the task itself - true before anyone plays it, and independent of what the camera can see.
MEASURED	What this session actually observed.	Only when the detector had the sensor data it needs, and the reading means what the detector thinks it means.

A game may target a reflex it cannot measure. It must never report a measurement it did not make.

Gate 1 - does the game have the sensors?

Every detector declares what it needs. `canMeasure()` checks that list against the game's sensor set, and a game that runs hands only fails the gate for every head- or face-based pattern. This is why **Babkin is shown as targeted but not measured in Pinch the Coin**: Babkin needs hand closure *and* mouth opening, the mouth needs Face Mesh, and Pinch the Coin runs Hands only.

Needs	Detectors	Available in
hands	Moro, Palmar Grasp	Every game
hands + head	ATNR, STNR, TLR	LetterQuest, Path Tracing
hands + face	Babkin, Hand-to-Mouth, Visual Tracking	LetterQuest, Path Tracing
face (+ head)	VOR, Eye Coordination	LetterQuest, Path Tracing
unavailable	Spinal Galant	No game - questionnaire only

Gate 2 - would the reading actually mean anything?

Having the sensor is not the same as the number being interpretable. The Palmar Grasp detector scores how much the hand pulls into a closed position during effort - which indicates a retained reflex **only when closing is not what the child was asked to do**.

The case that forced this second gate

In **Pinch the Coin**, **Finger Piano** and **Magic Finger Copy**, closing the hand is the task. The detector cannot tell a deliberate pinch from an involuntary grasp, so it would score a child playing the game correctly as having a strongly retained reflex - **the more skilful the child, the worse the report**.

Those three games therefore **target** Palmar Grasp (the activity genuinely works on it) but are flagged measurable: `false` so it is never scored. Where the task demands a sustained open or pointing hand - LetterQuest, Trace-Find-Type, Follow the Ladybug, Pop the Bubble, Path Tracing - closure is unwanted, so it means what the detector thinks it means.

Gate 3 - is there an honest sentence to write?

Every target carries a why string that is shown to parents and therapists, and it must say **what the child does** that loads the reflex - not what the reflex is. A reflex with no honest sentence of that kind does not belong in the list. This is why **Moro and TLR are absent from LetterQuest**: both are real and both were once reported there, but a seated spelling game has no startle stimulus and no whole-body posture change, so the activity does not train either one. They can still be measured there if they happen to occur - targets and measurements are different claims.

6 The number chosen for each game

Primary means the activity is substantially built around the pattern. Secondary means it loads it meaningfully but is not built on it.

Game	Sensors	N	Targets	Why this number
LetterQuest	hands face head	5	ATNR P , Palmar Grasp P , STNR P , Babkin S , Hand-to-Mouth S	The only game with the full sensor set, and a wide keyboard that forces repeated midline crossing with head turns. Moro and TLR are deliberately excluded - no startle, no posture change.
Path Tracing	hands face head	4	ATNR P , Palmar Grasp P , STNR S , Hand-to-Mouth S	Full sensors, and a continuous traced line that demands a sustained pointing hand across the midline.
Pop the Bubble	hands	3	ATNR P , Moro S , Palmar Grasp S	Hands only, but bubbles arrive on both sides and appear suddenly - the only hands-only game with a genuine, repeated startle-like stimulus, which is what earns Moro its place.
Trace - Find - Type	hands	2	Palmar Grasp P , ATNR S	Hands only, so the five face/head patterns are out. One finger must lead while the rest of the hand stays quiet, and the keyboard crosses the midline.
Follow the Ladybug	hands	2	ATNR P , Palmar Grasp S	Hands only. A path that wanders across the body is the ATNR load; the sustained point is the grasp load.
Pinch the Coin	hands	2	Palmar Grasp P (not measured), Babkin S (no face)	Hands only, and pinching is the task - so the grasp detector is blocked by Gate 2 and Babkin by Gate 1. Both are still honest targets.
Magic Finger Copy	hands	2	Palmar Grasp P (not measured), Babkin S (no face)	Same shape as Pinch the Coin: copying a hand posture means closing the hand deliberately.
Finger Piano	hands	2	Palmar Grasp P (not measured), Babkin S (no face)	Same shape again - pressing keys is deliberate finger closure.

P = primary target **S** = secondary target **red** = targeted but deliberately not scored, with the reason printed on the child's report.

So: why does a game contain just two primitive reflexes?

Because five of the eight need a face or head signal that a hands-only game does not have (Gate 1), one of the remaining three - Moro - needs a startle stimulus that only Pop the Bubble provides (Gate 3), and Spinal Galant is not observable by any camera at all.

That leaves **ATNR and Palmar Grasp** as the two patterns a hands-only game can honestly claim - which is exactly the pair you see on Trace-Find-Type and Follow the Ladybug. The number is small on purpose: it is what the sensors and the task can actually support.

7 What reaches the administrator export

The admin dashboard produces a single .xlsx workbook per learner (or per session). It contains up to six sheets, each a different unit of analysis.

File name pattern

NestureAI_Session_Export_<learner>_<suffix>_<YYYY-MM-DD>.xlsx

Sheet	One row per	Columns
Sessions Export	Session	A title block and metadata first - <i>Generated On, Learner Name, Learner Age, Diagnosis, Parent/Guardian</i> - then the table: Date, Duration (s), Accuracy, Difficulty, LPI Score, Perfect Grabs, Failed Grabs, Avg Response (ms), Smoothness, Fatigue.
Hand Positions	Landmark, per frame	Session ID, Recorded At, Point Index, X (Normalized), Y (Normalized), Timestamp (ms). Source table: raw_hand_tracking. This is the camera-mode raw data.
Pose Sessions	Pose session	Session ID, Game, Recorded (UTC), Duration (s), Frames, Sample Hz, Schema, Indicator model, Consent version, Consent given at. The index every other pose sheet joins on.
Pose Features	Pose session	Session ID, Game, then 24 derived measures: frameCount, durationSec, sampleRateHz, meanQuality, headYawRange/Sd, headPitchRange/Sd, meanExtensionLeft/Right, extensionAsymmetryMean/Sd, shoulderTiltMean/Sd, trunkLeanSd, posturalSwaySd, midlineCrossings, wristSpreadSd, and the correlation columns with their sample counts.
Pose Indicators	Indicator, per session	Session ID, Game, Indicator, Reflex, Evidence tier, Measured, Value (0-100), Confidence (0-1), Not measured because, Model version, Caveat. Long format, ready for a statistics package. Value is left blank, never 0 , when nothing was measured.
Pose Frames	Landmark, per frame	Session ID, Game, Frame, Time (ms), Landmark, Index, X, Y, Z, Visibility. The tall sheet: 21 landmarks x 15 Hz = 315 rows per second of play.

Two deliberate safeguards in the export

Truncation is visible, never silent. An Excel worksheet holds 1,048,576 rows. The frame sheet stops at a budget of **900,000** rows and writes a marked red row saying what was left out and how to get the rest. An analyst who does not know the data stops early would read the gap as the child having stopped moving.

An unmeasured indicator is blank, not zero. A zero in the Value column would read as "no pattern present" - a finding the session did not make. The reason is carried in its own column instead, and the caveat text travels with every row.

Gates a pose session must pass before indicators are produced

Gate	Value	Meaning
MIN_FRAMES	90	About 6 seconds at 15 Hz. Below this nothing is stable.
MIN_QUALITY	0.45	Mean landmark visibility across the session.
MIN_YAW_RANGE	0.22	Head-yaw travel needed before ATNR can be looked for at all.
MIN_PITCH_RANGE	0.18	Shoulder-widths of nose travel needed for STNR.
MIN_PAIRED_SAMPLES	45	Frames where both sides of a correlation exist.

8 Limits of the method

Stated plainly, because a report that hides them invites the wrong reading.

Limit	What it means in practice
Not a diagnosis	These are movement observations from a webcam during play. They describe patterns to guide practice. They do not identify a condition, and no clinical decision should rest on them alone.
Pose indicators are provisional	The touch-mode indicator model is explicitly versioned -provisional and every reflex weight in it is zero. It is a research channel for gathering data, not a scoring channel. The architecture is built so the formulas can be refitted against real data later without touching capture, export or storage.
One camera, no depth	The z coordinate from a single webcam is noisy. It is recorded for completeness, but no indicator in the platform depends on it.
Absence of evidence is not evidence	Every layer distinguishes <i>not measured</i> from <i>measured and settled</i> . An empty accumulator must never produce a perfect score - that fault existed and was removed deliberately in the scoring, the report and the export.
Sensors bound the claim	Six of the eight games run hands only, so at most three primitive reflexes are even observable in them, and in three of those six the grasp reading is blocked because closure is the task. The reports say which, and why, on the card itself.

Source of every figure in this document

Taxonomy, per-game targets and the two measurement gates: `src/services/reflexProfiles.js`

Detector rules and constants: `src/services/reflexEngine/reflexDetectors/*.js`

Windowing, score bands and confidence: `src/services/reflexEngine/index.js`, `outputFormatter.js`

Capture graphs and landmark counts: `src/hooks/useHandTracking.js`, `useMediaPipeTracking.js`, `useUpperBodyTracking.js`, `src/services/pose/poseLandmarks.js`

Pose indicator model and gates: `src/services/pose/indicatorModel.v1.js`

Export sheets and columns: `src/pages/AdminDashboard.jsx`, `src/services/pose/poseExport.js`